Watsonx Orchestrate Chat with Agents

REST API for communication with IBM Watsonx Orchestrate

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Overview

This asset provides a **FastAPI-based HTTP service** that enables communication with **IBM Watson Orchestrate (WxO)** through three main endpoints:

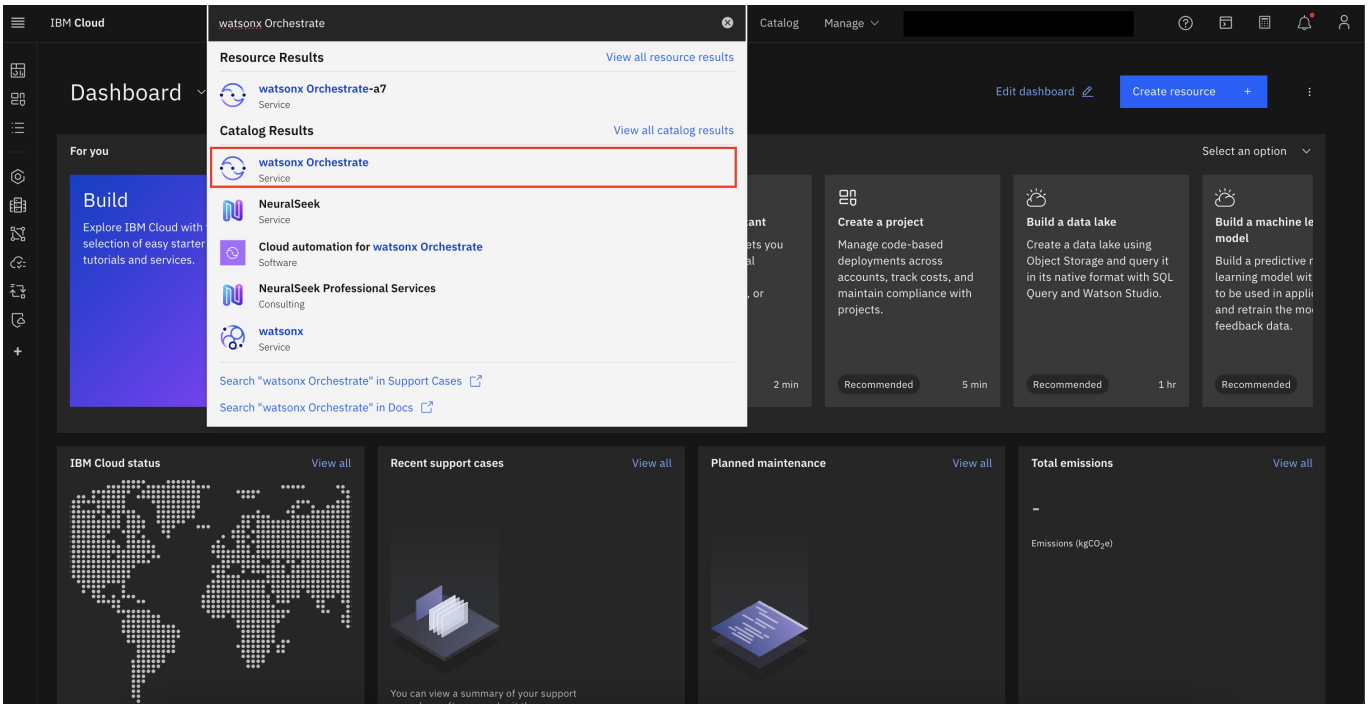
Endpoint	Description
`/list_agents`	Returns a list of available agents in a given watsonx Orchestrate instance
`/get_response`	Returns a single response from a WxO agent
`/stream_response`	Streams the agent's response in real-time using Server-Sent Events (SSE)

Useful Links

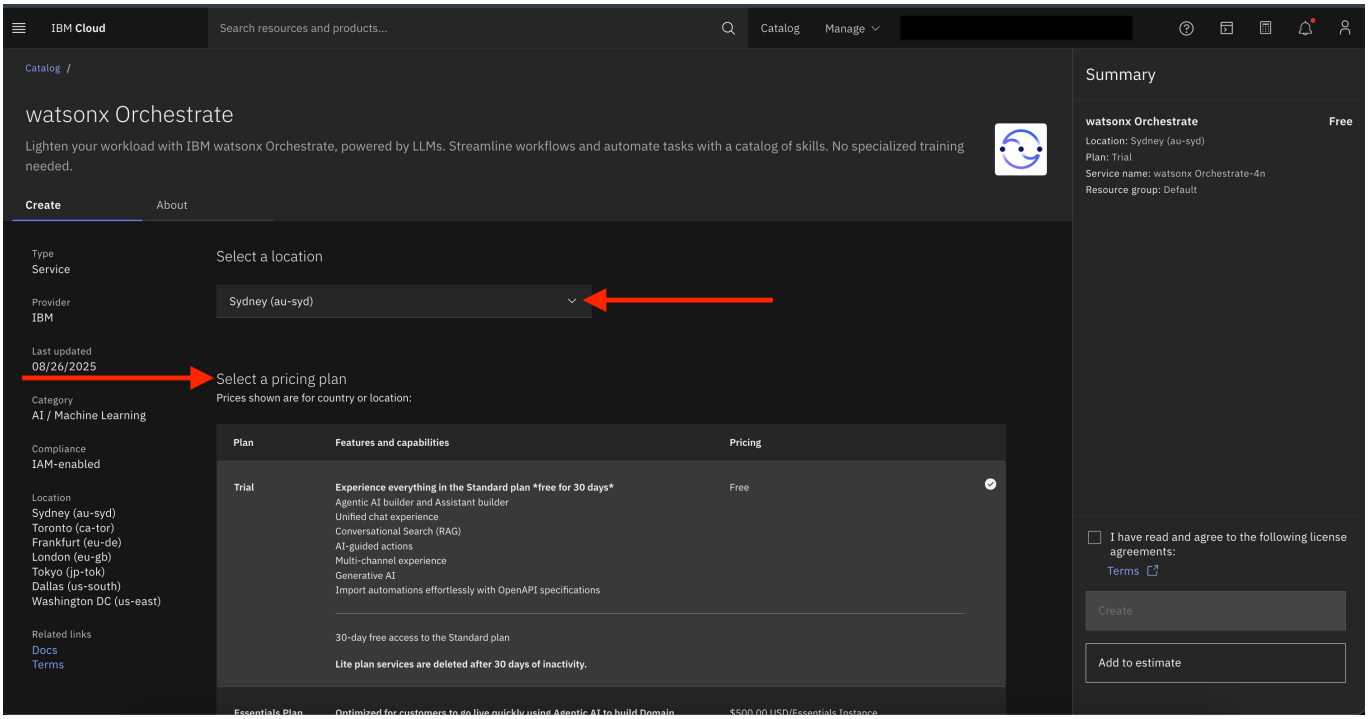
- **Source Code**: GitHub Repository
- **Implemented Endpoints**: Routes Directory

Create a Watsonx Orchestrate Instance

After logging into **IBM Cloud**, type `watsonx Orchestrate` in the search field.

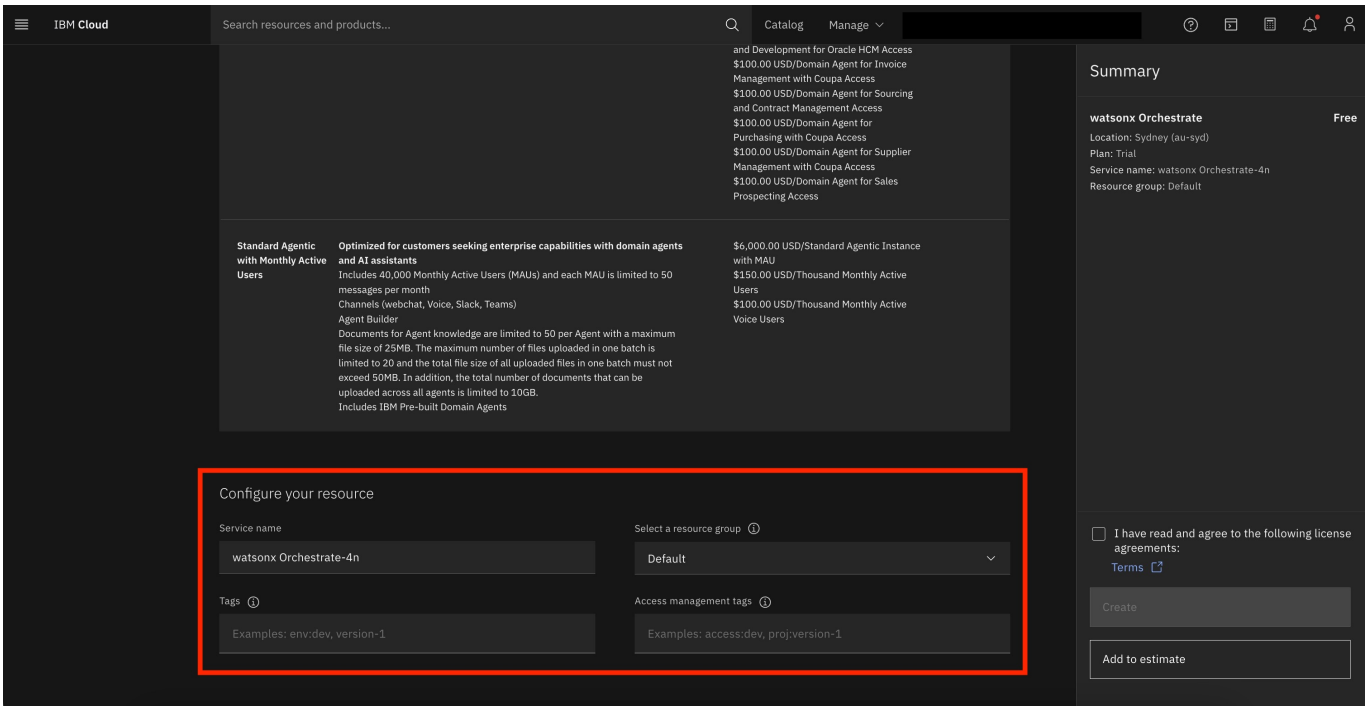


Choose the **location** for your instance and the **plan** that best suits your needs

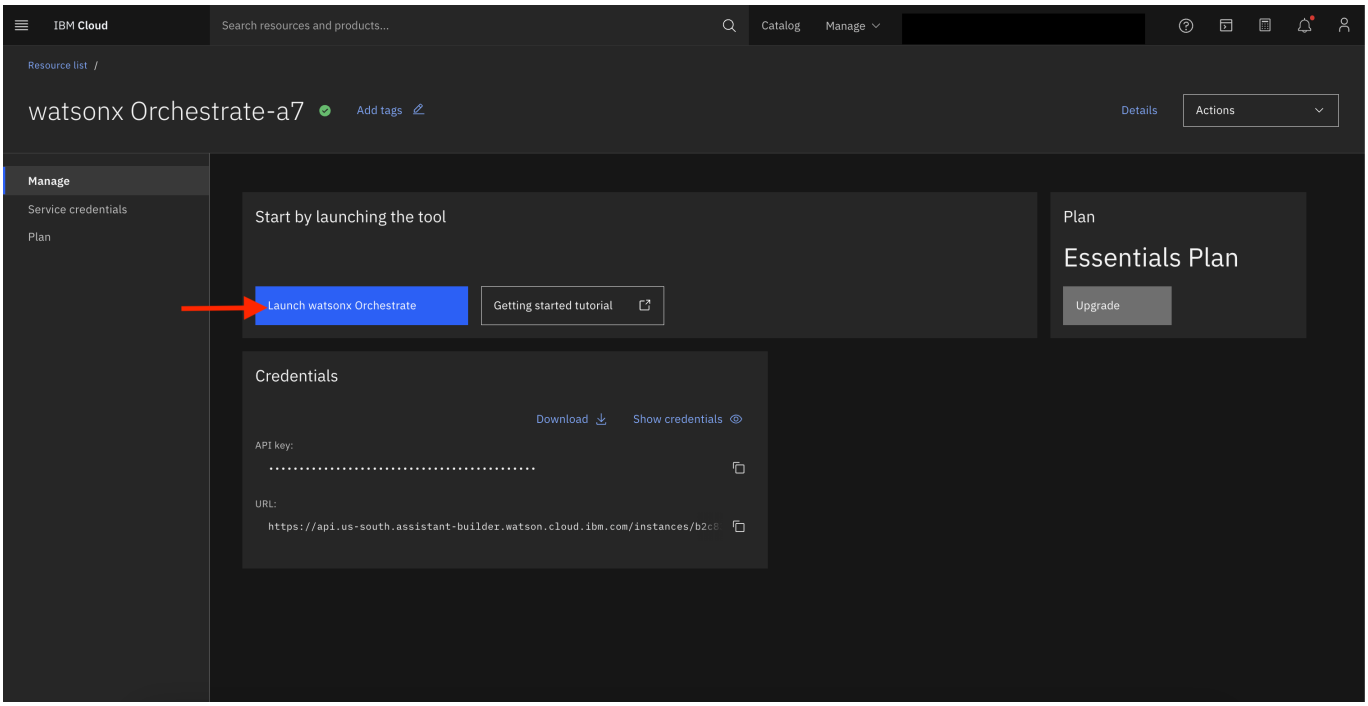


Finally, enter the **service name**, optionally **tags** and **access management tags**, and select a **resource group**.

Accept the IBM Cloud terms on the right side of the screen and click **Create**.



With your instance created, click **Launch watsonx Orchestrate**.

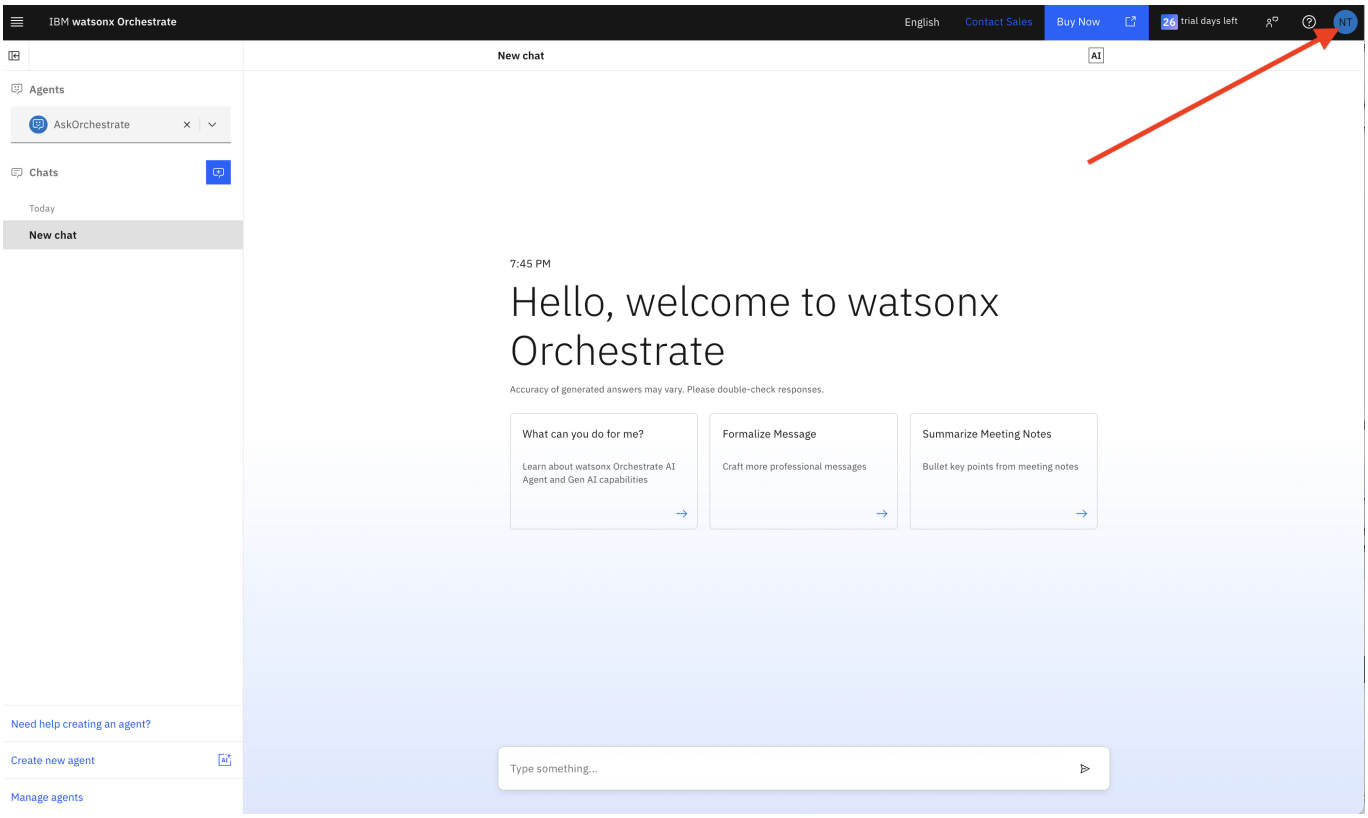


To communicate with Watsonx Orchestrate via API, you'll need to obtain two essential credentials:

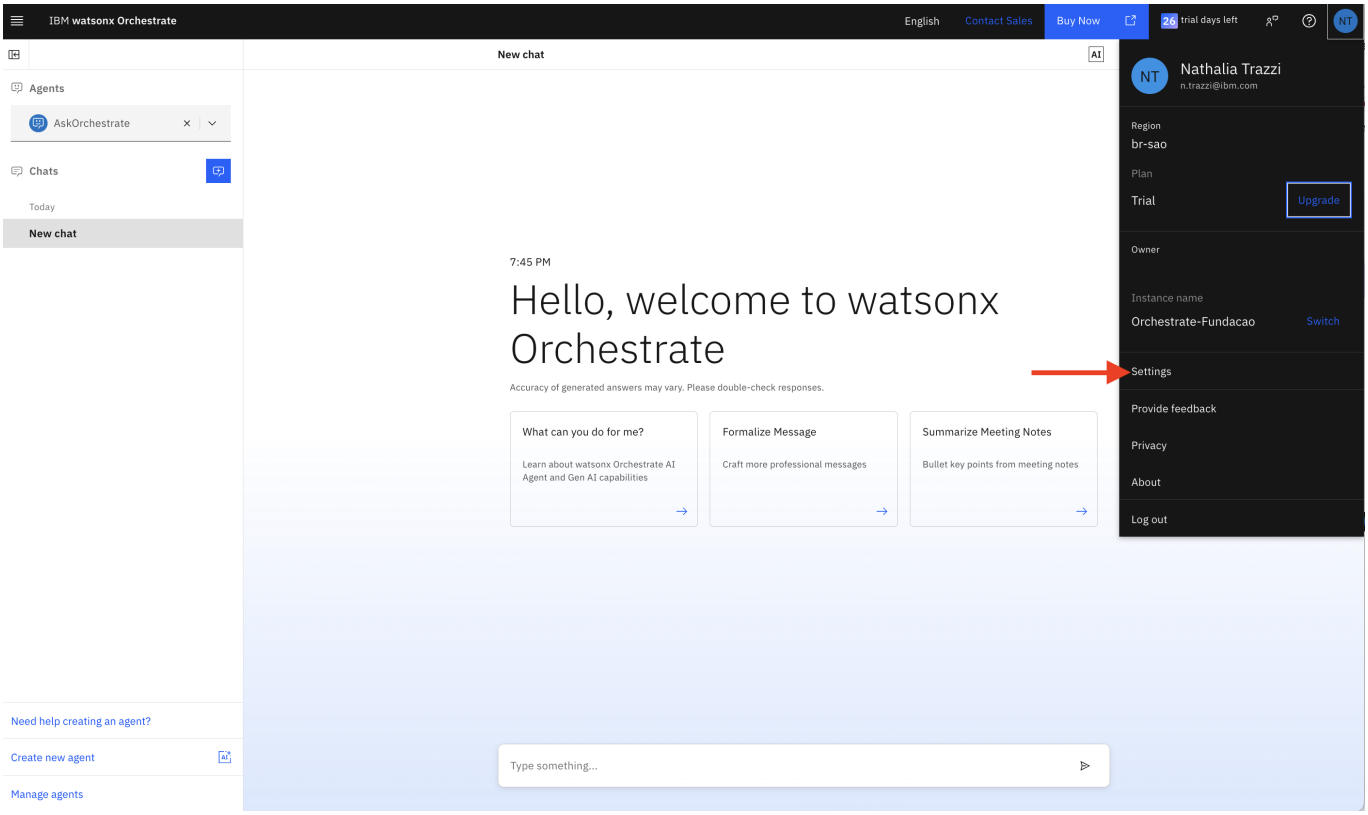
- `IAM_API_KEY`
- `WXO_INSTANCE_ID`

How to obtain watsonx Orchestrate credentials

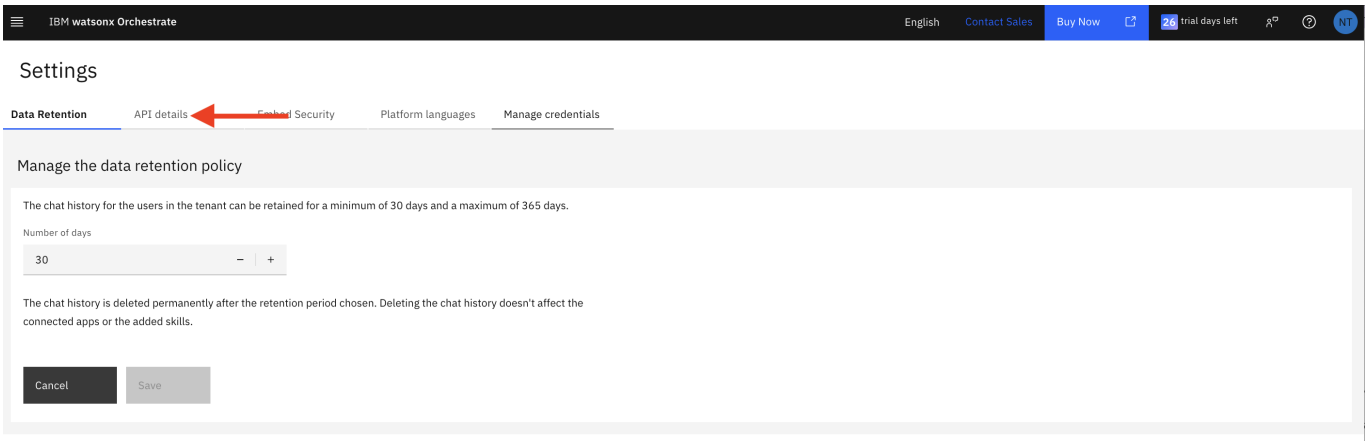
Click on the icon with your name and surname initials on the right side of the interface



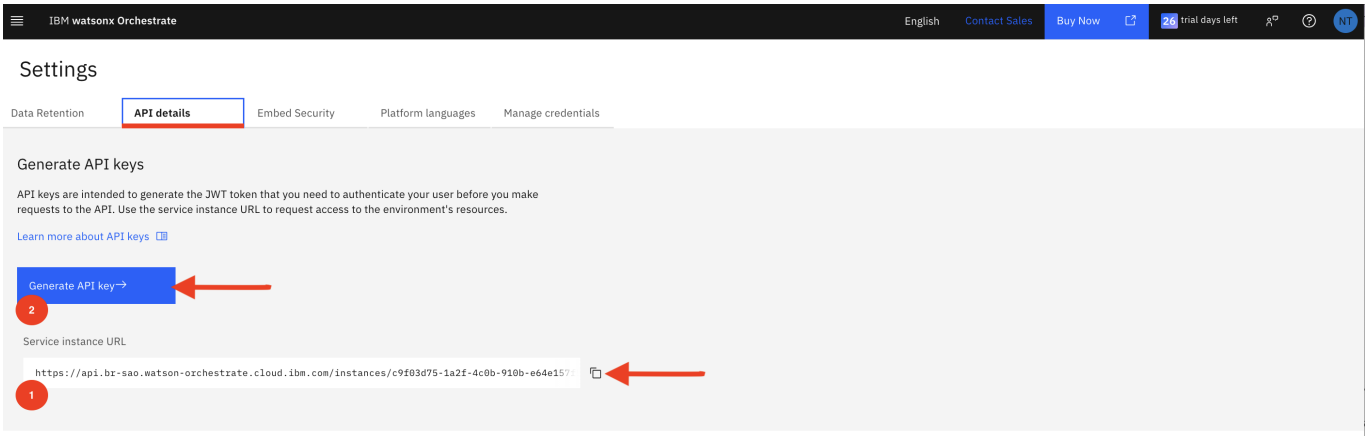
Click on **Settings**.



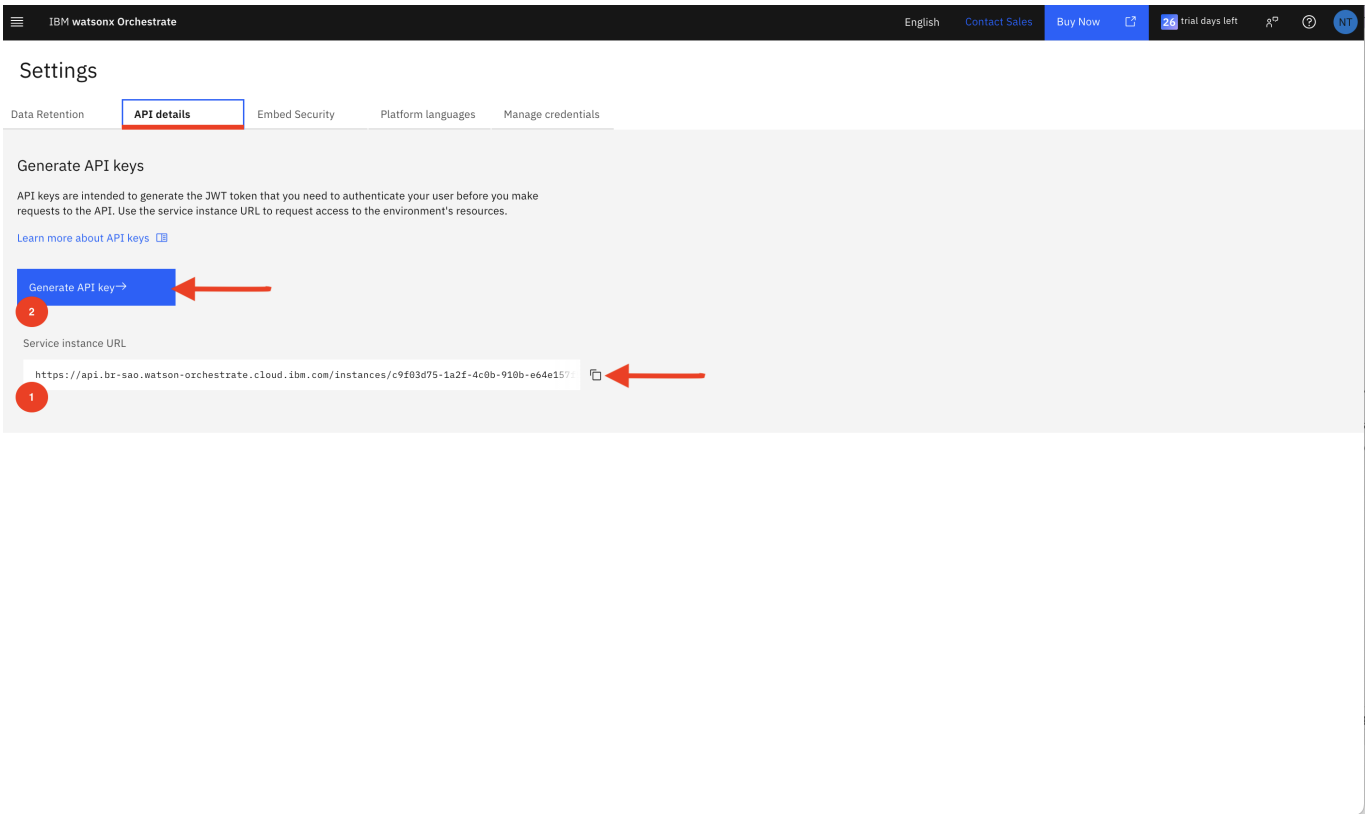
On the settings page, go to the **API details** tab



1. Copy your instance ID

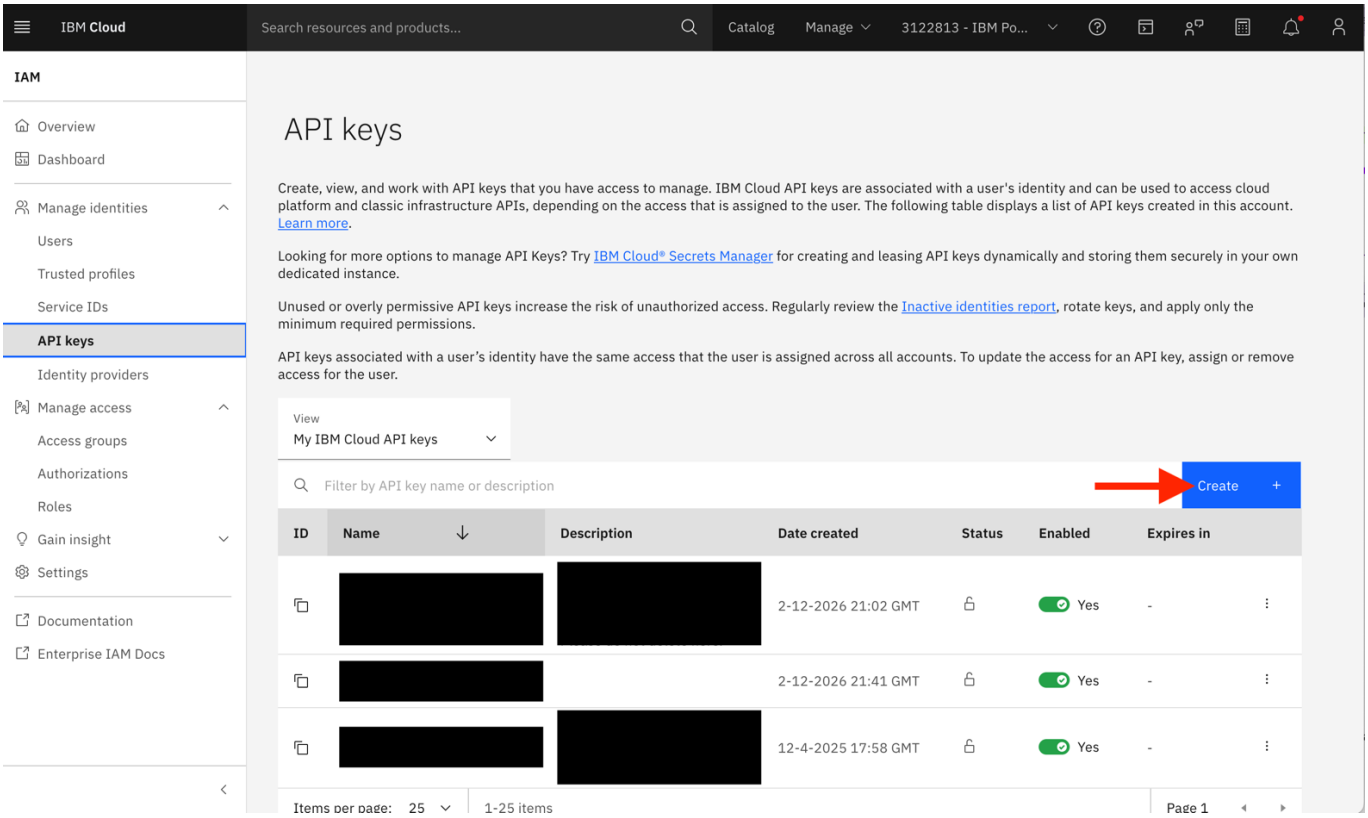


2. Next, click on `Generate API Key`



On the next page, click **Generate API key**.

You will be redirected to the catalog where you create, view, and work with IBM Cloud API keys.



> **What is an API Key?**
>
> An API Key is a secret that allows users, applications, or services to access IBM Cloud in an authenticated manner. It works as a "technical password" used in CLIs, automations, pipelines, or integrations.

API Key Configuration

Name

Give a descriptive name to the key, such as:

- `apikey-wxo`
- `local-development-key`

> **Tip**: Use names that indicate the purpose, responsible party, or environment.

Description (Optional)

Here you can add a description of your key, helping with its identification.

Lead Action (Action in Case of Leak)

Option	Description
🔘 **Disable the leaked key** (recommended)	The key is automatically disabled. Good option for production.
🔘 **Delete the leaked key**	The key is permanently deleted.
🔘 **Nothing**	No action is taken. Not recommended.

API Key Expiration

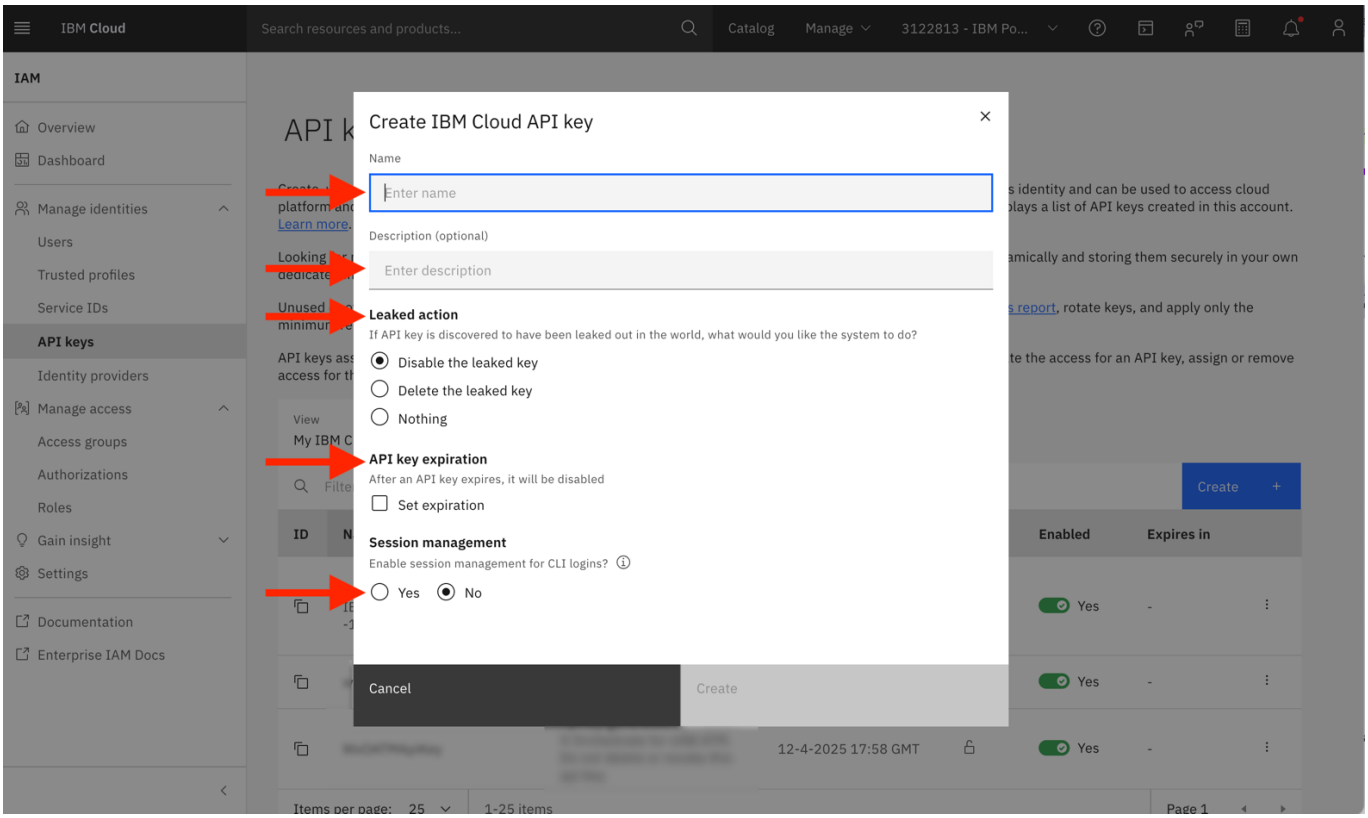
You can choose whether the key should expire automatically.

Option	Description
🔘 **Set expiration**	Allows you to choose a future date. -> Ideal for temporary access, short-term integrations, etc. When it expires, the key is automatically disabled.
🔘 **No expiration** (default)	The key does not expire.

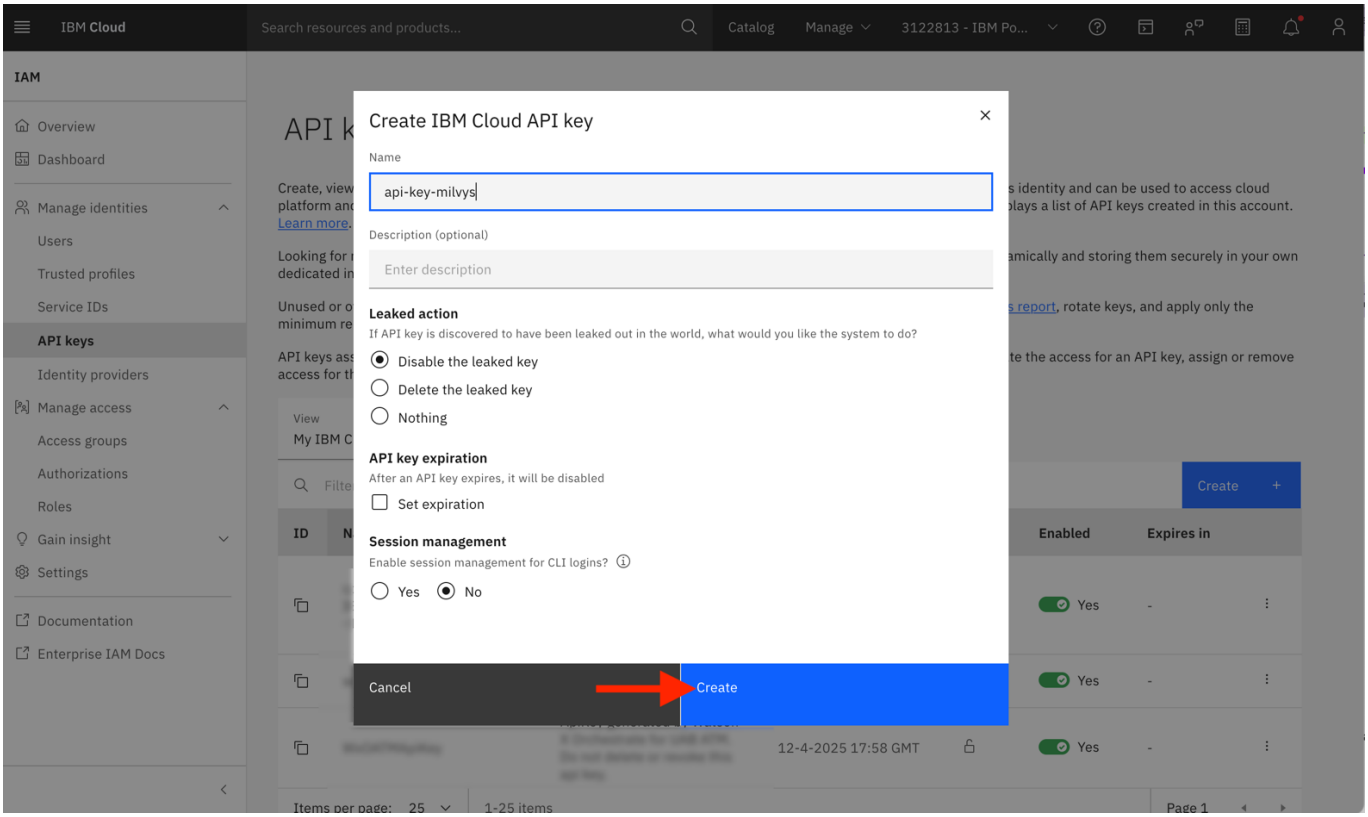
Session Management

Defines whether this key can be used for session management in IBM Cloud CLI.

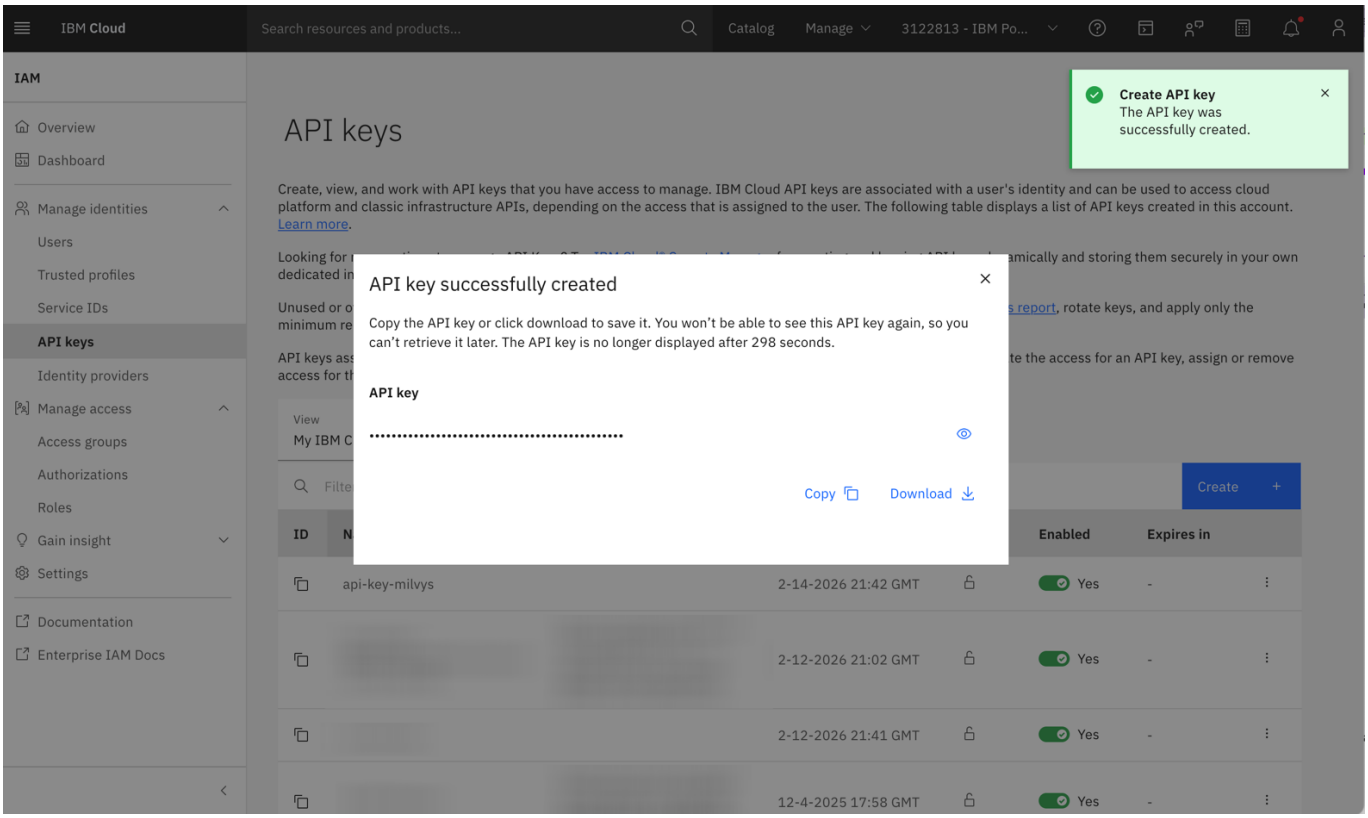
Option	Description
🔘 **Yes**	Allows the key to be used to log in to CLI with `ibmcloud login --apikey`
🔘 **No**	The API Key cannot be used to log in to CLI sessions. Use when you want to restrict the key only to specific API calls.



After filling everything out, click **Create**.



> **IMPORTANT**: Take this moment to copy or download your API Key. The key will be available for a few seconds. After this period, it will no longer be possible to view it again.



After obtaining the credentials, you now have the two pieces of information needed to execute backend interaction with your agent's chat (watsonx Orchestrate).

1. **Navigate to the project directory**:

```
cd ./source/rest_api/
```

2. **Configure environment variables**:

   - Rename the `.env_example` file to `.env`
   - Fill in with your credentials:

```
IAM_API_KEY=your-api-key-here
WXO_INSTANCE_ID=your-instance-id-here
```

3. **Install dependencies**:

```
pip install -r requirements.txt
```

4. **Run the server**:

```
python app.py
```

Conversation Management (Thread ID)

> **Important**: To maintain conversation context between multiple messages:

- **First message**: Make your first call to `/get_response` or `/stream_response` **without providing** a `thread_id` (leave empty or omit from the request). The response will include a `thread_id` value.
- **Subsequent messages**: To continue the conversation in the same context, include the returned `thread_id` as a parameter in all subsequent requests.

This allows the agent to maintain context between messages.

`POST /get_response`

Sends a message to an agent and receives a single response.

Request (query parameters):

```
POST /get_response?message=Hello&agent_id=your-agent-
id[&thread_id=optional-thread-id]
```

Parameters:

- `message` (required): The message to be sent
- `agent_id` (required): Agent ID
- `thread_id` (optional): Thread ID to continue a conversation

Notes:

- For the first message, **do not include** `thread_id`
- For subsequent messages, set `thread_id` with the value returned in the previous response

Response:

```
{
  "response": "Agent response here",
  "thread_id": "thread-id-value"
}
```

`POST /stream_response`

Sends a message to a WxO agent and receives the response continuously (SSE).

Request (query parameters):

```
POST /stream_response?message=Hello&agent_id=your-agent-id[&thread_id=optional-thread-id]
```

Parameters:

- `message` (required): The message to be sent
- `agent_id` (required): Agent ID
- `thread_id` (optional): Thread ID to continue a conversation

Notes:

- For the first message, **do not include** `thread_id`
- For subsequent messages, set `thread_id` with the value returned in the previous response

Response:

```
Content-Type: text/event-stream

data: {"chunk": "Response", "thread_id": "thread-id-value"}

data: {"chunk": " chunk", "thread_id": "thread-id-value"}

data: {"chunk": " here", "thread_id": "thread-id-value"}
```

Each event data chunk is sent in real-time as the agent processes the response.

Environment Variables

Variable	Description
`IAM_API_KEY`	IBM Cloud IAM API Key
`WXO_INSTANCE_ID`	Watsonx Orchestrate instance ID

Important Notes

> **Attention**:
>
> - Make sure your **agent ID** is valid
> - Your **IAM key** must have sufficient permissions to access the instance
> - The `/stream_response` endpoint uses **streaming (SSE)** to interact with the watsonx Orchestrate agent processing responses in real-time

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Now you're ready to integrate your application with IBM Watson Orchestrate.

Developed by Nathalia Trazzi